

Phase 1: Foundation

Chapter 1: Project Foundation

Problem Statement

Aaj adhiktar colleges hackathon organize karne ke liye bahut saare alag-alag tools ka use karte hain.

Example:

- Google Form → Registration
- WhatsApp → Announcements
- Gmail → Communication
- Google Drive → Project Submission
- Excel → Team Management
- Manual Sheets → Judging
- Canva/PDF → Certificates

Is wajah se:

- Data alag-alag jagah par hota hai.
- Organizer ka kaam mushkil ho jata hai.
- Participants confuse ho jate hain.
- Judges ko projects manage karna difficult hota hai.

Solution

Ek aisa web platform banana jahan hackathon ke saare kaam ek hi jagah se ho saken.

Is platform par:

- Organizer hackathon create karega.
- Participants register karenge.
- Teams banengi.
- Projects submit honge.
- Judges evaluation karenge.
- Leaderboard automatically banega.

Yaani **One Platform for Complete Hackathon Management.**

Users

1. Super Admin
2. Organizer
3. Participant
4. Team Leader
5. Team Member
6. Judge
7. Mentor (*Future*)

Golden Rule ★

Har feature kisi na kisi user ki problem solve karta hai.

Agar koi user hi nahi hai, to us feature ki zarurat bhi nahi hai.

Ye line poore project ki foundation hai.

Chapter 2: User Analysis

User 1: Organizer

Organizer kaun hai?

Woh person ya college jo hackathon organize karta hai.

Examples:

- College
- University
- Coding Club
- Technical Society
- Company

Organizer ka Main Goal

Main bina kisi confusion ke poora hackathon manage karna chahta hoon.

Ab socho...

Jab organizer pehli baar website open karega, to woh kya karega?

Uski journey likho.

Organizer Journey

1. Sign Up
2. Login
3. Organization Profile Create
4. Create New Hackathon
5. Fill Basic Details
6. Publish Hackathon
7. Receive Registrations
8. Manage Teams
9. Assign Judges
10. Start Evaluation
11. Announce Winners
12. Close Hackathon

Ye **Organizer User Flow** kehlata hai.

Ab har step ko todte hain.

Step 1: Create New Hackathon

Organizer ko kya-kya information bharni hogi?

Basic Information

- Hackathon Name
- Description
- Theme
- Mode (Online / Offline / Hybrid)
- Start Date
- End Date
- Registration Deadline
- Team Size (Minimum & Maximum)
- Prize Details
- Rules
- Contact Information
- Banner Image

Ye sab ek form ka hissa hoga.

Software Engineering Rule ★

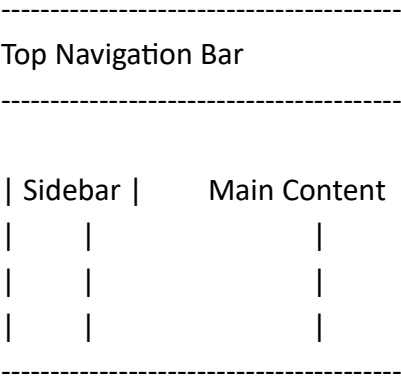
Har screen ka ek clear purpose hona chahiye.
Agar ek screen par 20 kaam honge, to user confuse ho jayega.
Isliye hum chhoti aur simple screens banayenge.

Chapter 3: Organizer Dashboard

Dashboard ka Purpose

Dashboard ek control center hota hai jahan se organizer poore hackathon ko manage karta hai.

Dashboard Layout



1. Top Navigation Bar

Isme honge:

- Platform Logo
- Search
- Notifications 
- Profile Photo
- Settings

2. Left Sidebar

Yeh sabse important part hai.

1. Dashboard
2. My Hackathons
3. Participants
4. Teams
5. Judges
6. Announcements
7. Project Submissions
8. Evaluation
9. Leaderboard
10. Certificates
11. Settings

 Dhyan do: Yeh features nahi hain. Yeh sirf navigation menu hai.

3. Main Dashboard

Organizer login karega to sabse pehle usse summary dikhni chahiye.

Likho:

Dashboard Cards

- Total Hackathons
- Total Participants
- Total Teams
- Total Judges
- Projects Submitted
- Registration Status (Open/Closed)

Ye cards organizer ko ek nazar me platform ki condition bata denge.

Recent Activity

Ek section hoga jahan latest updates dikhenge.

Example:

- Team Alpha registered.
- Judge Rahul accepted invitation.
- Team CodeX submitted project.
- Registration closed.
- Winner announced.

Quick Actions

Dashboard par kuch bade buttons honge.

-  Create New Hackathon
-  Add Judge
-  Make Announcement
-  Export Report

Software Engineering Rule

Dashboard information dikhata hai.

Management alag pages par hoti hai.

Example:

- Dashboard → "500 participants"
- Participants Page → Un 500 participants ko manage karna.

Ye difference bahut important hai.

Chapter 4: Platform Architecture

Sabse Pehla User – Super Admin

Super Admin kaun hai?

Super Admin platform ka owner hota hai.

Example:

- Agar platform ka naam **HackSphere** hai, to uska owner tum hoge.
- Colleges owner nahi honge, woh sirf platform ke customers/users honge.

Super Admin ki Responsibilities

- Platform ko manage karna
- Organizers ko approve karna (optional)
- Sabhi hackathons dekhna
- Sabhi users dekhna
- Reports dekhna
- Platform settings manage karna
- Platform security manage karna

Ab ek bahut important concept ✨

Real World Example

Socho...

Instagram ka owner kaun hai?

Meta

Kya har Instagram user Meta ban jata hai?

Nahi.

Waise hi...

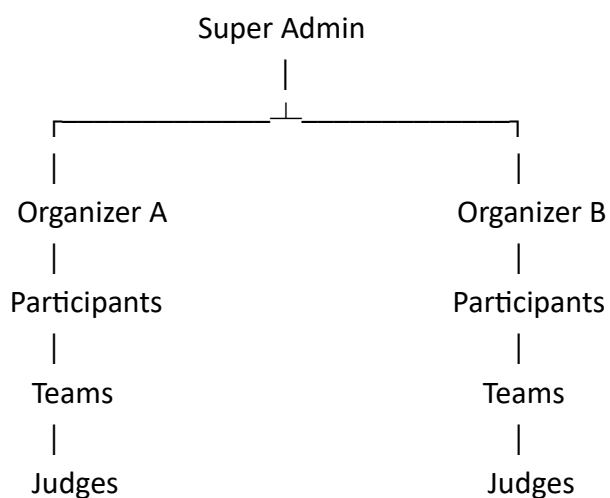
Tumhara platform hoga.

Us par 100 colleges account bana sakte hain.

Lekin platform ka owner sirf tum hoge.

Yahi Super Admin hai.

User Hierarchy



Ye diagram batata hai ki platform mein authority kis order mein hai.

Golden Rule ★

Har user ke paas sirf utni hi permission hogi jitni uske role ke liye zaruri hai.

Is concept ko **Role-Based Access Control (RBAC)** kehte hain. Ye almost har professional software mein use hota hai.

Chapter 5: Entities (Platform ki Foundation)

"Ek software data se banta hai."

Aur data kis cheez ka hota hai?

Entities ka.

Agar platform me sirf ek organizer hai to uske paas kya-kya hoga?

1. Organization

Information:

- Organization Name
- Logo
- Email
- Phone
- Website
- Address
- Description

2. Hackathon

Information:

- Name
- Theme
- Description
- Start Date
- End Date
- Registration Date
- Team Size
- Mode
- Rules
- Prize Pool

3. Participant

Information:

- Name
- Email
- College
- Phone
- Skill

4. Team

Information:

- Team Name
- Team Leader
- Members
- Hackathon
- Status

5. Judge

Information:

- Name
- Email
- Expertise
- Organization

6. Submission

Information:

- GitHub Link
- Demo Video
- PPT
- Description
- Submitted Time

7. Evaluation

Information:

- Judge
- Team
- Score
- Feedback

8. Announcement

Information:

- Title
- Message
- Date

Ab sabse important part ★

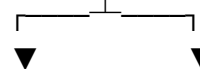
In sab entities ka relation kya hai?

Ye diagram banao:

Organization



Hackathon



Team

Announcement



Submission



Evaluation

Participant —————▶ Team

Judge —————▶ Evaluation

Ab ek software engineering concept.

Parent aur Child Relationship

- Ek Organization ke bahut saare Hackathons ho sakte hain.
- Ek Hackathon me bahut saari Teams ho sakti hain.
- Ek Team ek Submission karegi.
- Ek Submission ko kai Judges evaluate kar sakte hain.

Isse hum **relationship** bolte hain.

Golden Rule ★

Features badal sakte hain.

UI badal sakta hai.

Lekin data model strong hona chahiye.

Isi wajah se companies pehle database aur relationships sochti hain, baad me UI banati hain.

Chapter 6: Relationships

Rule 1

One Participant → Many Hackathons

Example:

Shivam

- AI Hackathon 2026 
- Smart India Hackathon 
- Web Dev Challenge 
- CodeFest 2027 

Ek hi account.

Kai hackathons.

Rule 2

One Team belongs to only One Hackathon

Example:

AI Hackathon

→ Team Alpha

→ Team CodeX

→ Team Vision

Ye teams kisi dusre hackathon me use nahi hongi.

Rule 3

One Participant can join many Teams (at different times)

Example:

2026

AI Hackathon

→ Team Alpha

2027

Web Challenge

→ Team Vision

Participant wahi hai.

Team alag hai.

Hackathon alag hai.

Rule 4

One Judge can judge many Teams

Example:

Judge Rahul

→ Team Alpha

→ Team Vision

→ Team Phoenix

Rule 5

One Organization can host many Hackathons

Example:

Jagannath University

→ Hackathon 2026

→ Hackathon 2027

→ AI Challenge

→ Cyber Security Cup

Ab sabse important concept

Real Identity vs Event Identity

Participant ki **Real Identity** kya hai?

- Name
- Email
- Phone

Ye kabhi change nahi honge (ya bahut kam honge).

Lekin **Event Identity** har hackathon me alag hogi.

Example:

Hackathon A

- Team Alpha
- Team Leader

Hackathon B

- Team Vision
- Team Member

Yani **same participant, different role in different hackathons.**

Ye concept bahut powerful hai.

Chapter 7: Team Management

Principle

Organizer hackathon manage karega.

Participants apni team khud manage karenge.

Participant Journey

1. Sign Up
2. Login
3. Browse Hackathons
4. Register for Hackathon
5. Create Team **ya**
6. Join Existing Team
7. Invite Members
8. Submit Project
9. Track Results

Team Creation Flow

Participant → Register → Create Team

Uske baad system:

- Team ID generate karega.
- Team Leader assign karega.
- Invite Code ya Invite Link generate karega.

Baaki members us code/link se join karenge.

 **Organizer ko is process me involve hone ki zarurat nahi hogi.**

Team Leader ki Permissions

- Team Name change kar sakta hai.
- Members ko invite kar sakta hai.
- Request accept/reject kar sakta hai.
- Project submit kar sakta hai.
- Team details edit kar sakta hai.

Team Member ki Permissions

- Team join kar sakta hai.
- Team details dekh sakta hai.
- Project files upload kar sakta hai (*agar leader allow kare*).
- Team leave kar sakta hai.

Golden Rule ★

Jis user ka kaam hai, wahi user woh kaam kare.

Doosre user par unnecessary responsibility mat daalo.

Ye UX (User Experience) ka bahut bada principle hai.

Chapter 8: Authentication & Authorization

Authentication batata hai "Aap kaun ho?"

Authorization batata hai "Aap kya kar sakte ho?"

Ye dono alag cheezein hain.

Authentication

Ye sirf identity verify karta hai.

Example:

- Login
- Signup
- Forgot Password
- OTP
- Email Verification

Ye sab Authentication ka part hain.

Authorization

Login ke baad system decide karega ki user kya kar sakta hai.

Example:

👤 Organizer

- Hackathon create kar sakta hai. ✅
- Judge assign kar sakta hai. ✅
- Kisi aur organization ka hackathon edit nahi kar sakta. ❌

👤 Participant

- Team bana sakta hai. ✅
- Project submit kar sakta hai. ✅
- Hackathon delete nahi kar sakta. ❌

 Judge

- Assigned projects evaluate kar sakta hai. 
- Winner declare nahi kar sakta. 

Ab sabse important decision

Ek Email = Ek Account

Platform me ek email se sirf ek account banega.

Example:

shivam@gmail.com

Ye email dubara account create nahi kar sakta.

Lekin...

Is ek account ke paas future me multiple roles ho sakte hain.

Real Life Example

Socho...

Rahul ek college student hai.

Aaj:

- Participant hai.

Do saal baad:

- Judge ban gaya.

5 saal baad:

- Organizer ban gaya.

Kya Rahul ko har baar naya account banana padega?

Nahi.

Uska wahi account rahega.

Bas uske roles badhte jayenge.

Isliye likho

User aur Role alag cheezein hain.

User

|

└— Participant

└— Judge

└— Organizer

└— Super Admin

Ek user ke paas ek ya ek se zyada roles ho sakte hain.

Ye design future-proof hota hai.

Login Flow

User Login

|



Authentication

|



System checks Role

|

└— Super Admin → Admin Dashboard

└— Organizer → Organizer Dashboard

└— Judge → Judge Dashboard

└— Participant → Participant Dashboard

Golden Rule ★

User ko login ke baad wahi dikhna chahiye jo uske role ke hisaab se zaruri hai.

Chapter 9: Organization-Centric Architecture

Platform kisi college ke liye nahi, kisi bhi organization ke liye bana hai.

Organization kya ho sakti hai?

- University
- College
- School
- Coding Club
- Startup

- Company
- NGO
- Government Organization
- Community
- Individual Organizer (Future)

Yani jo bhi hackathon organize karna chahe, woh platform use kar sake.

Organization Profile

Har organization ke paas apna profile hoga.

Information:

- Organization Name
- Logo
- Description
- Website
- Email
- Phone
- Address
- Social Links
- Verified Status

Organization Dashboard

Har organization sirf apna data dekhegi.

Example:

Jagannath University

→ Sirf apne hackathons.

Microsoft

→ Sirf apne hackathons.

Google

→ Sirf apne hackathons.

Koi bhi organization dusri organization ka data nahi dekh sakti.

Golden Rule ★

Data isolation is security.

Matlab har organization ka data alag aur secure rahega.

Ab sabse important concept

Workspace

Ye concept Notion, Slack aur GitHub jaise products me bhi use hota hai.

Har organization ka ek **workspace** hoga.

Us workspace ke andar:

- Hackathons
- Members
- Judges
- Participants
- Announcements
- Certificates

Sab kuch usi organization ka hoga.

Isse future me ek organization ke multiple admins bhi ho sakte hain.

Chapter 10: Public Hackathon Marketplace

Platform ek Hackathon Marketplace hoga.

Yaani platform par jitne bhi public hackathons honge, participant un sab ko explore kar sakta hai.

Participant Home Page

Participant login karega to usse ye sections dikh sakte hain:

-  Featured Hackathons
-  Newly Added
-  Registration Closing Soon
-  Online Hackathons
-  College Hackathons
-  Company Hackathons
-  Recommended for You (*Future*)

Search & Filters

Participant search kar sake:

- Hackathon Name
- Organization Name
- Theme (AI, Web, Cyber Security, IoT...)
- Mode (Online/Offline/Hybrid)
- Prize Pool
- Registration Status
- Team Size

Hackathon Details Page

Har hackathon open karne par:

- Banner
- Organization
- Description
- Timeline
- Rules
- Eligibility
- Prize Pool
- Judges
- FAQs
- Register Button

Registration Flow

Explore Hackathons



Open Details



Click Register



Accept Rules



Register Successfully



Create Team / Join Team

Dhyan do ki team registration se pehle hackathon registration hoga.

Ek bahut important concept

Public vs Private Hackathon

Har hackathon do types ka ho sakta hai.

Public

- Sab dekh sakte hain.
- Koi bhi eligible participant register kar sakta hai.

Private

- Sirf invite ya secret code se access.
- Public listing me nahi dikhega.

Example:

- College Internal Hackathon
- Company Internal Hackathon

Golden Rule ★

Har hackathon by default Public ya Private me se ek hoga.

Ye setting organizer hackathon create karte waqt choose karega.

Bhai... 🙌🙌 Ye answer sirf sahi nahi hai, **isme product thinking bhi hai.**

Sabse achhi baat ye hai ki tumne sirf "Draft hona chahiye" nahi bola, **uska reason bhi diya.** Product managers isi tarah decisions lete hain.

Apni copy me ye likho.

Chapter 11: Hackathon Lifecycle

Har hackathon Draft se shuru hoga.

Hackathon create karte hi woh public nahi hoga.

Lifecycle

Create Hackathon



Draft



Complete Details



Preview



Publish



Registration Open



Registration Closed



Hackathon Running



Judging



Completed



Archived

Draft Mode

Organizer Draft me ye sab kar sakta hai:

- Details edit
- Banner change
- Rules update
- Judges add
- FAQs add
- Prize pool change
- Timeline change

Participants ko Draft hackathon **kabhi nahi dikhega**.

Preview Mode

Ye bahut important feature hai.

Publish karne se pehle organizer dekh sakega:

Participant ko hackathon exactly kaisa dikhega?

Agar sab sahi lage...

Tab hi Publish.

Publish Rules

Publish button tabhi active hoga jab mandatory fields complete hon.

Example:

- ✓ Hackathon Name
- ✓ Banner
- ✓ Registration Date
- ✓ Start Date
- ✓ End Date
- ✓ Description
- ✓ Rules

Agar kuch missing ho...

✗ Publish disabled.

Golden Rule ★

Incomplete information should never reach the public.

Ye sirf hackathon ke liye nahi, poore platform ke liye rule hoga.

Chapter 12: Change Management

Har information ka impact alag hota hai.

Isliye har field ko same permission nahi milni chahiye.

Category 1: Safe Changes ✅

Ye changes registration ke baad bhi allow ho sakte hain.

- Banner
- Description
- FAQs
- Contact Details
- Sponsor Information
- Gallery

Ye changes participants ko affect nahi karte.

Category 2: Sensitive Changes ⚠️

Ye changes participants ko directly affect karte hain.

Example:

- Registration Deadline
- Start Date
- End Date
- Prize Pool
- Team Size
- Eligibility

Inhe change kiya ja sakta hai, **lekin system ko sabhi registered participants ko notification bhejni chahiye.**

Example:

"Hackathon Update: Maximum team size has been changed from 4 to 5 members."

Isse transparency bani rahegi.

Category 3: Locked Changes

Ye publish hone ke baad change nahi hone chahiye.

- Organization
- Hackathon ID
- Hackathon Type
- Theme (*optional, agar theme competition ka core hai*)
- Basic Structure

Ye change karna participants ka trust tod sakta hai.

Golden Rule

Platform ka pehla kaam organizer ko power dena nahi, balki sabhi users ke saath fairness maintain karna hai.

Chapter 13: Database Design (Blueprint)

Database software ka dil (Heart) hota hai.

UI badal sakta hai.

Backend badal sakta hai.

Programming language badal sakti hai.

Lekin agar database galat design hua, to poora software future me problems dega.

Rule No. 1

Ek table = Ek entity

Example:

 Users

 Organizations

 Hackathons

 Teams

 Submissions

 Judges

Ye sab alag tables honge.

Sabse pehla table

USERS

user_id

full_name

email

phone

password

profile_photo

created_at
updated_at
status

Question 🤔

Yahan **role** kyu nahi likha?

Kyuki humne pehle hi decide kiya tha...

👉 Ek user future me multiple roles rakh sakta hai.

Isliye role ko alag table me rakhenge.

Second Table

Roles

role_id
role_name

Data kuch aisa hoga:

role_id	role_name
1	Super Admin
2	Organizer
3	Participant
4	Judge

Third Table

User_Roles

Ye table bahut important hai.

user_role_id
user_id
role_id

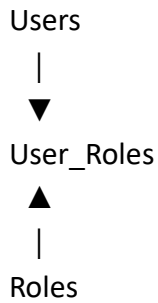
Example:

Rahul

Participant ✅

Judge ✅

Dono roles isi table me store honge.



Ab ye kyu kiya?

Agar kal ko hume ek naya role add karna ho.

Example:

Mentor

Volunteer

Sponsor

To kya Users table badalna padega?

✗ Nahi.

Sirf Roles table me ek nayi row add hogi.

Isi ko kehte hain...

Scalable Database Design

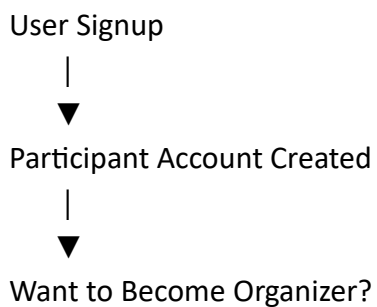
Golden Rule ★

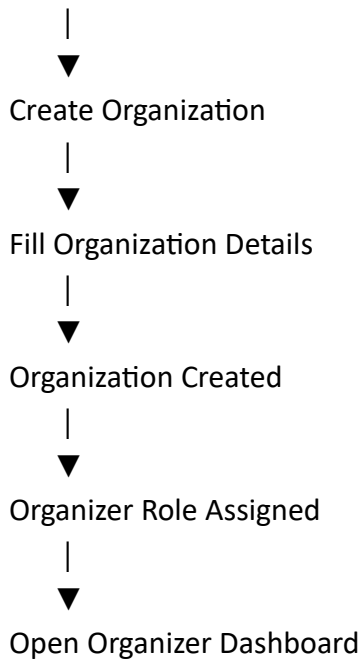
Kabhi bhi future ko dhyan me rakhkar database design karo.

Chapter 14: Organization Onboarding

Organization pehle banti hai, Organizer baad me banta hai.

Complete Flow





Organization Creation Form

Basic Details

- Organization Name
- Organization Type
- Logo
- Official Email
- Phone Number
- Website (Optional)
- Description
- Country
- State
- City

Organization Types

- University
- College
- School
- Company
- Startup
- Coding Club
- Community
- NGO
- Government
- Other

Organization Owner

Jis user ne organization banayi hogi:

- Wahi pehla Organization Owner hoga.
- Wahi pehla Organizer hoga.

Future me woh aur organizers add kar sakta hai.

Golden Rule ★

Ek platform account aur ek organization account alag cheezein hain.

Example:

Tumhare paas ek personal account hai.

Us account se tum:

- Participant ban sakte ho.
- Judge ban sakte ho.
- Organizer ban sakte ho.
- Aur future me do alag organizations bhi manage kar sakte ho.

Yahi professional design hota hai.

Chapter 15: Organization Ownership

Organization kisi ek insaan se badi hoti hai.

Agar owner platform chhod de, to organization band nahi honi chahiye.

Ownership Transfer

Current Owner



Select New Owner



New Owner Accepts Request



Ownership Transferred



Old Owner becomes Organizer/Admin

Organization Roles

1. Owner

- Full control
- Transfer ownership
- Delete organization
- Manage billing (Future)
- Add/Remove admins

2. Admin (Organizer)

- Create hackathons
- Manage hackathons
- Manage judges
- Manage participants
- Publish announcements

3. Staff (Future)

- Limited permissions
- View reports
- Help in management

Golden Rule ★

Owner is also an Organizer, but every Organizer is not the Owner.

Chapter 16: Permission System

Har organizer ko same permission nahi milegi.

Example:

Owner

✓ Sab kuch

Admin

✓ Hackathon create

✓ Publish

✓ Judges manage

✗ Ownership transfer

✗ Organization delete

Event Manager

- ✓ Participants manage
- ✓ Teams manage
- ✓ Announcements
- ✗ Organization settings

Judge Coordinator

- ✓ Judges assign
- ✓ Evaluation manage
- ✗ Hackathon delete

Professional Principle

Role batata hai user ka designation.

Permission batati hai user kya kar sakta hai.

Ye dono alag concepts hain.

Chapter 17: Rules Engine

Ek achha software har kaam ke liye insaan par depend nahi karta.

Matlab jitne kaam automatically ho sakte hain, unhe system khud kare.

Hackathon Timeline

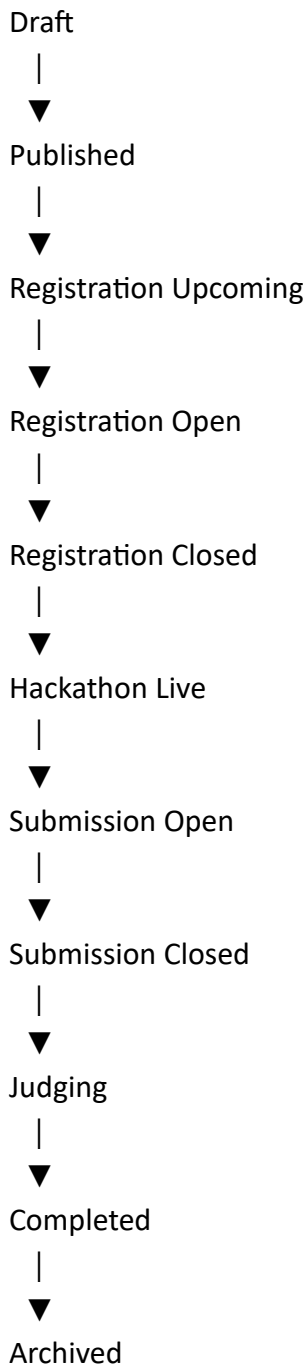
Har hackathon ke paas ye dates hongi.

- Registration Start
- Registration End
- Hackathon Start
- Submission Start
- Submission End
- Judging Start
- Judging End
- Result Date

Ye sirf dates nahi hain.

Ye poore system ko control karenge.

State Machine



Ye states manually change nahi hongi.

System date aur time dekhkar khud state change karega.

Example

Maan lo:

Registration Start: 1 August 2026

Registration End: 10 August 2026

Aaj date hai: 5 August

System automatically dikhayega: ☒ Registration Open

11 August

System automatically

✖ Register Button hide

Registration Closed

Organizer ko kuch karne ki zarurat nahi.

Automatic Rules

Registration Open → Register Button ON

Registration Closed → Register Button OFF

Submission Open → Submit Button ON

Submission Closed → Submit Button OFF

Judging → Sirf Judges access

Completed → Result Visible

Archive → Read Only

Notifications

Har state change par notification.

Example:

Registration Open → "Registration has started."

Registration Close hone se 24 ghante pehle → "Only 24 hours left."

Submission Close hone se 1 ghanta pehle → Reminder

Result Publish → Congratulations 🎉

Golden Rule ★

Time is the most trusted manager.

Isliye jitne decisions date aur time se ho sakte hain, unhe system khud handle kare.

Aur ab... 🧠

Main tumhe ek real-world situation deta hoon.

Hackathon me submission ki last date hai:

10 August, **11:59 PM**

Ek team **11:59:58 PM** par submit button dabati hai.

Lekin internet slow hai aur file upload hote-hote **12:00:07 AM** ho jata hai.

Ab system kya kare?